

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



Dr. Swati Rajput

Assistant Professor

Department of Electronics Engineering

Indian Institute of Technology (Indian School of Mines) Dhanbad

Digital Electronics: Course Goals and Key Takeaways



Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance



Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:	Reference Books:
1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.	1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015. 2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Digital Electronics: Lecture Plan



Sl. No.	Topics	No of classes (tentative)
1.	Introduction to Number Systems and Conversion Techniques	8
2.	Continue.....	
3.	1's and 2's complement	
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	
5.	Boolean Algebra: Laws and Theorems	
6.	Boolean Function Implementation Using NAND/NOR Gates	
7.	Determination of Boolean Functions from Digital Circuits	
8.	Continue.....	
9.	SOP Standard Forms of Logical Functions	8
10.	POS Standard Forms of Logical Functions	
11.	Simplification of Logical Functions Using Boolean Algebra	
12.	Introduction to Karnaugh Maps (K-Maps)	
13.	K-Map Simplification (2-variable Functions)	
14.	K-Map Simplification (3-variable Functions)	
15.	K-Map Simplification (4-variable Functions)	
16.	Don't Care Conditions in K-Map Simplification	

Digital Electronics: Lecture Plan



17.	Introduction to Combinational Logic Circuits	8
18.	Design of Adders (Half Adder, Full Adder)	
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	
20.	Magnitude Comparator Design	
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	
22.	Multiplexers: Design and Applications	
23.	Demultiplexers: Design and Applications	
24.	Decoders and Encoders	
25.	Introduction to Sequential Circuits	10
26.	Latches: SR Latch, D Latch	
27.	Flip-Flops: SR, JK Flip-Flops	
28.	Flip-Flops: D and T Flip-Flops	
29.	Asynchronous Counters: Design and Operation	
30.	Continue.....	
31.	Synchronous Counters: Design and Operation	
32.	Continue.....	
33.	Shift Registers: Types and Applications	
34.	Continue.....	

Digital Electronics: Weightage



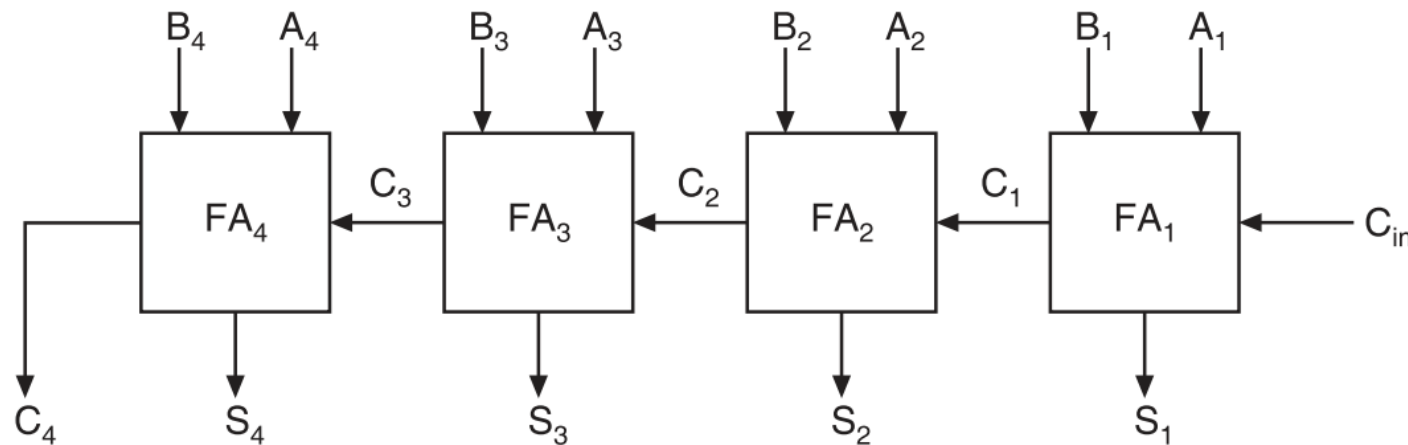
Weightage of Different Components

1.	Quiz/ Assignment	10 Marks
2.	Mid Semester Examination	30 Marks
3.	Quiz/ Assignment	10 Marks
4.	End Semester Examination	50 Marks
Total Marks = 100		

Binary Parallel Adder



- ❖ A binary parallel adder is a digital circuit that adds two binary numbers in parallel form and produces the arithmetic sum of those numbers in parallel form.
- ❖ It consists of full adders connected in a chain, with the output carry from each full-adder connected to the input carry of the next full adder in the chain.
- ❖ The interconnection of four full-adder (FA) circuits to provide a 4-bit parallel adder.
- ❖ The augend bits of A and addend bits of B are designated by subscript numbers from right to left, with subscript 1 denoting the lower-order bit.
- ❖ The carries are connected in a chain through the full-adders.
- ❖ The input carry to the adder is C_{in} and the output carry is C_4 . The S outputs generate the required sum bits.



Logic diagram of a 4-bit binary parallel adder.

An n -bit parallel adder requires n -full adders.

Number System



(r-1)'s Complement

Example: Obtain 2's complement of 10111010.

Answer: $N = 10111010$; $r = 2$; $n = 8$

(r-1)'s complement = $r^n - 1 - N$

(2-1)'s complement = $11111111 - 10111010 = 01000101$

2's complement = 1's complement + 1 = $0101 + 1 = 01000110$

Problem: Obtain 1's complement of 100111

Answer: 011000

Problem: Obtain 2's complement 1100111

Answer: 0011001

Binary Parallel Subtractor



Problem 1 : Subtract $(0101)_2$ from $(1011)_2$ using 2's complement

Conventional Method:

```
 1011
- 0101
-----
 0110
```

Complement's Method:

- Binary subtraction can be performed conveniently using **complements**.
- The subtraction $A - B$ is carried out by taking the **2's complement of B** and adding it to A.
- The **2's complement** of a binary number is obtained by:
 - ✓ Taking the **1's complement** of the number
 - ✓ Adding **1** to the least significant bit (LSB)

$B = (0101)_2$

(a) **1's complement** (invert bits)

$0101 \rightarrow 1010$

(b) **Add 1** $\rightarrow 1010 + 1 = 1011$

Step 3: Add to the minuend A

```
  1011
+ 1011
-----
 10110
```

Step 4: Discard the carry

The leftmost **carry** is discarded. **Answer is 0110**

Binary Parallel Subtractor



Problem 1 : Perform the subtraction with the following unsigned binary numbers (a) by taking the 2's complement of the subtrahend and (b) by taking the 1's complement of the subtrahend.

(i) 11010 – 10000

$$\begin{array}{rcll} \text{(i)} & 11010 & \Rightarrow & 11010 \\ & -10000 & & +10000 \quad \text{(2's complement)} \\ & & & \hline & & & \textcircled{1}01010 \quad \text{(Ignore the carry)} \\ & & & \hline & = +01010 & = +10 \quad \text{(Result is positive = +10)} \end{array}$$

There is a carry. Ignore it. The MSB is 0. Hence, the answer is positive and is in true binary form.

So it is $+01010 = +10$

Binary Parallel Subtractor



Problem 1 : Perform the subtraction with the following unsigned binary numbers (a) by taking the 2's complement of the subtrahend and (b) by taking the 1's complement of the subtrahend.

(ii) 11010 – 1101

$$\begin{array}{rcll} \text{(ii)} & 11010 & \Rightarrow & 11010 & = 26 \\ - & 1101 & & + 10011 & \text{(2's complement)} & = -13 \\ & & & \hline & & & \textcircled{1}01101 & \text{(Ignore the carry, result is positive = +13)} & +13 \\ & & & \hline & & & = +01101 = +13 \end{array}$$

There is a carry. Ignore it. The MSB is 0. Hence, the answer is positive and is in true binary form.

So it is $+01101 = +13$

Binary Parallel Subtractor



Problem 1 : Perform the subtraction with the following unsigned binary numbers (a) by taking the 2's complement of the subtrahend and (b) by taking the 1's complement of the subtrahend.

(iii) 100 – 11000

$$\begin{array}{rcll} \text{(iii)} & 100 & \Rightarrow & 0000100 \\ - & 110000 & \Rightarrow & 0110000 \\ & & \Rightarrow & + \quad 0000100 \\ & & & + \quad 1010000 \quad (2's \text{ complement}) \\ & & & \hline & & & 1010100 \quad (\text{Result is negative}) \\ & & & \hline & & & = -0101100 = -44 \end{array} \quad \begin{array}{r} = 4 \\ = -48 \\ = -44 \end{array}$$

There is no carry. The MSB is 1. Hence, the answer is negative. Take its 2's complement and put a minus sign. So, it is $-101100 = -44$

Binary Parallel Subtractor



Problem 1 : Perform the subtraction with the following unsigned binary numbers (a) by taking the 2's complement of the subtrahend and (b) by taking the 1's complement of the subtrahend.

(iv) 1010100 – 1010100

$$\begin{array}{rcll} \text{(iv)} & 1010100 & \Rightarrow & 1010100 \\ & - 1010100 & & + 1010100 \quad (2's \text{ complement}) \\ & & & \hline & & & \text{1} 0000000 \quad (\text{Ignore the carry}) \\ & & & = 0 \quad (\text{Result} = 0) \end{array} \qquad \begin{array}{r} = 84 \\ = -84 \\ \hline 0 \end{array}$$

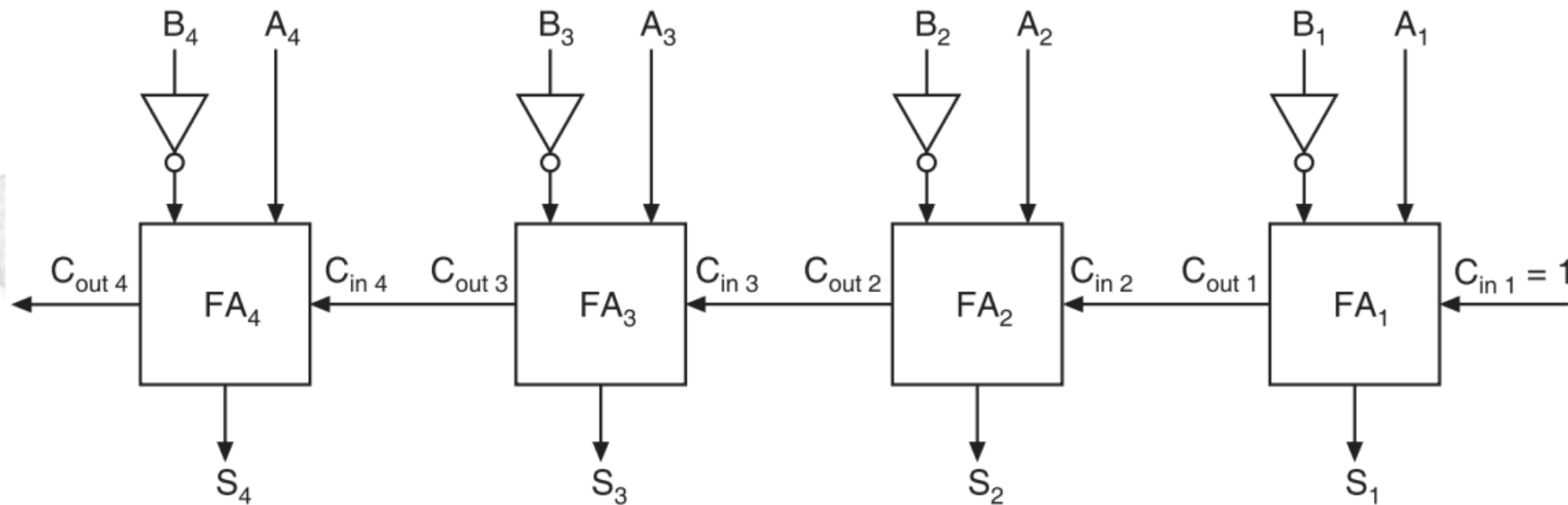
There is a carry. Ignore it. The MSB is 0. Hence, the answer is positive and is in true binary form.

So it is $+000000 = 0$.

Binary Parallel Subtractor



- Binary subtraction can be performed conveniently using **complements**.
- The subtraction $A - B$ is carried out by taking the **2's complement of B** and adding it to A.
- The **2's complement** of a binary number is obtained by:
 - Taking the **1's complement** of the number
 - Adding **1** to the least significant bit (LSB)
- The **1's complement** is formed by **inverting all bits** (changing 0 to 1 and 1 to 0).
- The 1's complement operation can be implemented using **inverters**, as illustrated in the figure.



Logic Diagram of a 4-bit Binary Adder Subtractor

Binary Adder-Subtractor



- A **4-bit adder-subtractor circuit** combines both addition and subtraction in a single circuit.
- The circuit uses **one common binary adder** for both operations.
- An **X-OR gate** is included with each full adder to enable operation control.
- A **mode input (M)** controls the function of the circuit.
- When **M = 0**, the circuit operates as an **adder**.
- When **M = 1**, the circuit operates as a **subtractor**.
- Each X-OR gate receives:
 - ✓ The mode input **M**
 - ✓ One bit of input **B**
- When **M = 0**:
 - ✓ $B \oplus 0 = B$
 - ✓ The input carry is **0**
 - ✓ The circuit performs the operation **A + B**
- When **M = 1**:
 - ✓ $B \oplus 1 = B'$
 - ✓ The input carry **C1** is set to **1**
 - ✓ The **B inputs are complemented**
 - ✓ A **1 is added** through the input carry
 - ✓ The circuit performs **A + (2's complement of B)**